

Solution Requirements

ComicCraft - AI Comic Story Creator using Gemini Models
Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester
Project period: 1 Oct 2026 - 2 Oct 2026

Functional requirements

ID	Requirement
FR1	Web form accepts prompt, character, setting, tone and style
FR2	Create a 5-panel outline as structured JSON
FR3	Write caption and narration for each panel
FR4	Generate one image per panel
FR5	Show a preview of all panels
FR6	Export the comic as a PDF
FR7	Offer a JSON API at /generate-comic/json
FR8	Offer /test-image for image-only checks

Non-functional requirements

ID	Requirement
NFR1	Secrets only in .env, excluded from Git
NFR2	Image model loaded once, warmed up at startup
NFR3	Failures return a clear HTTP 500 message
NFR4	Responsive UI for desktop and mobile
NFR5	PDF supports Latin Unicode via DejaVu fonts
NFR6	Runs on a 16 GB Apple Silicon laptop (fp16, attention slicing)

Constraints

- Gemini free tier: 5 requests/min and 20 requests/day per model
- Local image generation needs several GB of disk and memory
- google-generativeai SDK is deprecated but used as specified